

NURNBERG, GermanyWMO: **107630**Lat: **49.5031N**Lon: **11.0550E**Elev: **1042**StdP: **14.15**Time Zone: **1.00 (EUC)**Period: **03-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	12.3	17.7	5.2	7.5	16.0	11.9	10.5	21.0	27.6	47.9	23.8	42.4	4.6	10	0.513

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	20.0	89.4	67.3	85.6	66.2	82.0	64.9	69.1	83.0	67.7	80.7	66.5	78.4	6.7	130

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
64.8	95.3	72.1	64.0	92.7	71.5	62.5	87.8	70.4	33.7	83.8	32.6	80.7	31.5	78.4	76.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 20.3	16.9	14.4	DB	7.8	94.7	6.6	2.6	3.0	96.6	-0.8	98.2	-4.6	99.6	-9.4	101.6	(4)
(5)			WB	7.2	72.1	6.6	1.9	2.4	73.4	-1.4	74.5	-5.1	75.6	-9.9	77.0	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	49.9	33.0	33.9	41.1	50.5	56.8	64.2	67.5	65.5	58.6	50.0	41.4	35.7
	DBStd	13.99	8.01	7.86	7.55	7.16	7.02	6.61	6.15	6.22	5.95	6.83	6.52	7.27
	HDD50	2162	527	452	283	80	20	1	0	0	3	87	266	443
	HDD65	5802	991	872	740	434	267	94	46	72	207	465	707	907
	CDD50	2139	0	0	8	96	231	425	542	479	262	87	9	0
	CDD65	308	0	0	0	1	13	69	123	86	16	0	0	0
	CDH74	3413	0	0	0	42	197	791	1299	901	175	8	0	0
	CDH80	1120	0	0	0	5	42	248	484	310	31	0	0	0
Wind	WSAvg	6.0	7.0	6.5	6.8	5.8	6.2	5.5	5.8	5.2	5.0	5.5	6.3	6.8
Precipitation	PrecAvg	24.5	1.7	1.4	1.7	1.6	2.5	2.7	2.9	2.5	2.0	1.9	1.9	2.0
	PrecMax	35.8	4.3	4.4	7.0	3.8	5.9	5.7	5.6	6.9	4.6	6.1	4.9	5.3
	PrecMin	15.7	0.4	0.2	0.1	0.3	0.5	0.6	0.1	0.1	0.4	0.1	0.0	0.2
	PrecStd	4.2	0.8	0.9	1.1	0.9	1.2	1.2	1.3	1.2	1.0	1.2	0.8	1.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	53.9	56.9	66.6	78.7	84.6	91.6	95.0	92.9	84.2	75.0	61.1	53.7
		MCWB	49.2	45.9	50.9	57.8	64.2	68.1	68.9	68.1	64.7	60.6	55.1	49.2
	2%	DB	50.2	51.5	62.5	73.7	79.1	86.2	89.8	87.7	79.0	68.3	57.5	50.3
		MCWB	46.2	45.5	49.5	54.9	62.1	66.2	67.2	67.4	63.5	58.8	52.2	47.1
	5%	DB	46.7	48.1	57.6	70.0	75.1	82.3	85.7	82.7	75.0	64.4	54.0	48.0
		MCWB	44.1	43.5	47.6	54.2	59.7	64.9	66.1	66.3	62.2	57.1	49.9	45.4
	10%	DB	44.2	44.7	53.7	66.0	71.3	78.6	80.9	78.8	71.2	60.9	51.5	45.0
		MCWB	42.2	41.3	46.7	52.8	58.0	63.6	64.8	64.6	60.8	55.6	48.1	43.2
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	50.3	48.5	53.4	59.7	66.0	70.3	70.1	71.1	67.3	62.5	56.1	50.6
		MCDB	53.4	52.0	61.4	75.2	80.1	84.6	85.0	84.0	77.7	70.1	60.1	53.1
	2%	WB	47.1	45.9	51.1	57.0	63.7	68.1	68.7	69.3	65.2	59.9	53.2	48.2
		MCDB	49.5	50.5	59.2	70.4	76.1	81.3	83.3	82.7	75.4	66.2	57.1	50.5
	5%	WB	44.2	43.5	49.1	55.0	61.3	66.2	67.5	67.5	63.3	58.0	50.6	45.2
		MCDB	46.5	47.2	56.4	67.6	72.2	78.7	81.2	79.3	72.9	63.2	54.1	47.2
	10%	WB	41.6	41.1	46.9	53.4	59.1	64.6	66.5	65.7	61.5	56.1	48.4	43.2
		MCDB	43.3	44.1	53.1	64.1	68.6	76.4	78.5	76.4	69.3	60.7	51.2	44.9
Mean Daily Temperature Range	5% DB	MDBR	9.4	11.6	16.6	20.5	19.2	19.9	20.0	19.6	18.9	15.9	10.6	9.1
		MCDBR	12.2	17.5	26.1	28.7	27.0	27.3	28.9	27.4	26.0	21.7	15.3	12.2
		MCWBR	9.7	11.8	14.8	13.7	12.4	11.0	9.8	10.4	12.1	12.7	10.8	9.5
	5% WB	MCDBR	11.8	15.5	22.1	25.8	23.9	24.1	22.9	23.9	21.9	19.4	13.8	11.1
MCWBR		9.7	11.1	13.5	13.0	11.7	10.3	8.8	10.2	11.0	11.9	10.3	9.2	
Clear-Sky Solar Irradiance	taub	0.306	0.329	0.365	0.398	0.405	0.407	0.416	0.410	0.380	0.356	0.331	0.303	
	taud	2.428	2.372	2.317	2.256	2.267	2.290	2.261	2.301	2.363	2.420	2.449	2.427	
	Ebn at Noon	237	258	266	268	271	271	266	262	258	244	225	221	
	Edh at Noon	20	27	34	39	41	40	41	37	32	25	20	18	
All-Sky Solar Radiation	RadAvg	269	532	879	1347	1592	1764	1669	1449	1055	627	304	213	
	RadStd	47	91	111	178	181	143	160	138	141	75	38	24	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47)	Regional (43 neighbors)	N/A	N/A	N/A	N/A	N/A	+0.72	N/A	N/A	N/A	N/A

Nomenclature: See separate page